

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

# International Islamic University Chittagong



Department of Computer Science & Engineering (CSE)

## Project Proposal

|                     |                                                 |
|---------------------|-------------------------------------------------|
| <b>Course Code</b>  | CSE-3532                                        |
| <b>Course Title</b> | Tools and Technologies for Internet Programming |
| <b>Credit Hour</b>  | 2                                               |
| <b>Semester</b>     | Spring 2026                                     |

| Submitted By             |                                                                                                                       | Submitted To    |                     |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------|---------------------|
| Name of Students         | 1. Minhaj Hasan Rohan (ID:C241101)<br>2. Mohammad Sadman Tahiat (ID:C241100)<br>3. Mehedi Hasan Howlader (ID:C241086) | Name of Teacher | Ahasanul Kalam Akib |
| Section                  | 5CM                                                                                                                   | Designation     | Adjunct Lecturer    |
| Semester                 | 5 <sup>th</sup>                                                                                                       | Department      | CSE                 |
| Date of Final Submission | 24/05/2026                                                                                                            |                 |                     |

Remark

Signature of the Teacher

## Project Title

### **Student Task Manager System-A Web Application for Academic Task Organization and Productivity**

## Problem Statement

University students today face an increasingly demanding academic workload. Balancing multiple courses, assignment deadlines, lab reports, presentations, and personal responsibilities makes effective task management not just helpful, but essential. Yet, most students still rely on informal methods — scattered paper notes, generic reminder apps, or simple memory — to manage their study tasks.

These informal approaches consistently fail in structured academic environments. Students frequently miss deadlines due to lack of prioritization, struggle to track the progress of ongoing tasks, and have no centralized system to visualize workload distribution across their semester.

***"67% of university students report struggling to organize academic tasks effectively, leading to missed deadlines and lower performance outcomes." — This project directly addresses that gap with a practical, digital solution.***

The Student Task Manager System is proposed as a purpose-built web application that gives students a centralized, intelligent, and intuitive platform to create, manage, prioritize, and track all their academic tasks — from a single dashboard accessible on any device.

Further challenges that justify the need for this system include:

- Absence of task prioritization based on deadlines and importance.
- No real-time overview of weekly or monthly academic workload.
- Difficulty in tracking progress of multiple ongoing assignments simultaneously.
- Over-dependence on manual reminders and inconsistent planning methods.
- No integrated system for organizing tasks, notes, and deadlines in one place.

## Objectives

The primary objective is to design and develop a fully functional, user-friendly, and scalable web-based task management system using the MERN stack (MongoDB, Express.js, React.js, Node.js) that addresses the real-world productivity needs of university students. Specific objectives include:

- Build a secure multi-user authentication system using JWT and bcryptjs for protected access.
- Enable students to create, update, prioritize, and delete academic tasks through an intuitive dashboard interface.
- Implement real-time task filtering and search capabilities for fast task retrieval.

- Provide a visual representation of task status (Pending / In Progress / Completed) for effective progress tracking.
- Deliver a responsive and accessible UI using React.js and Tailwind CSS, optimized for desktop and mobile browsers.
- Implement a RESTful API backend using Node.js and Express.js following industry-standard design principles.
- Deploy the application on cloud platforms (Vercel + Render + MongoDB Atlas) making it accessible 24/7.

## Key Features

| Feature           | Description                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------|
| Authentication    | Secure user registration and login using JWT and bcryptjs for protected access.               |
| Task Management   | Create, edit, delete, and update tasks with title, description, deadline, and priority level. |
| Task Status       | Toggle tasks between Pending, In Progress, and Completed status.                              |
| Search & Filter   | Live search by title; filter by status and sort by deadline or creation date.                 |
| Dashboard Summary | Visual summary cards: total, completed, pending, and overdue tasks.                           |
| Dark/Light Mode   | User-selectable theme for comfortable use in any environment.                                 |
| Responsive Design | Fully functional across desktop, tablet, and mobile screen sizes.                             |

### 4.1 Core Authentication

**User Registration** — Students can create a personal account with name, email, and a securely hashed password.

**User Login & JWT Authentication** — Login session is maintained via JSON Web Tokens stored securely in the browser.

**Protected Routes** — All task-related routes are protected; unauthenticated users are redirected to login.

### 4.2 Task Management

**Create Task** — Add tasks with title, description, due date/deadline, and priority level (Low / Medium / High).

**View All Tasks** — Tasks are displayed in an organized card/table layout on the main dashboard.

**Edit Task** — Modify any field of an existing task at any time.

**Delete Task** — Permanently remove tasks that are no longer needed.

**Mark as Complete** — Toggle tasks between Pending, In Progress, and Completed status with one click.

### 4.3 Productivity & UX Features

**Search Tasks** — Instantly search tasks by title using a live search bar.

**Filter & Sort** — Filter by status (Completed / Pending / High Priority) and sort by deadline or creation date.

**Dashboard Summary** — Visual summary cards showing total, completed, pending, and overdue tasks.

**Dark / Light Mode Toggle** — User-selectable theme for comfortable use in any environment.

**Responsive Design** — Fully functional across desktop, tablet, and mobile screen sizes.

## Technology Stack

| Category           | Technologies Used                            |
|--------------------|----------------------------------------------|
| Frontend           | HTML5, CSS3, Tailwind CSS, JavaScript (ES6+) |
| Backend            | Node.js, Express.js                          |
| Database           | MongoDB / MySQL                              |
| Authentication     | Firebase Authentication                      |
| API Architecture   | RESTful API                                  |
| Charts & Analytics | Chart.js                                     |
| Development Tools  | Visual Studio Code, GitHub (Version Control) |

## Target Users

The primary target audience for this system is:

- **University Students** — The core users. Any student managing course assignments, lab reports, project deadlines, and exam preparation.
- **College Students** — Applicable to any academic institution where task and deadline management is required.
- **Self-Learners & Professionals** — Anyone engaged in structured learning programs or personal project management.

*The system is designed to be universally accessible via any modern web browser, requiring no installation. This ensures the lowest possible barrier to adoption across all user groups.*

### Expected Outcome

Upon successful completion, Student Task Manager System will deliver the following benefits:

| Outcome                     | Description                                                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------|
| Easy Task Management        | Users can create, categorize, and review tasks with minimal effort.                                                |
| Deadline Awareness          | Automated deadline reminders help students stay on top of upcoming submissions and assignments.                    |
| Progress Tracking           | Visual status indicators (Pending / In Progress / Completed) help students understand and manage their workload.   |
| Better Academic Performance | Dashboard analytics and priority-based filtering empower students to make smarter, more organized study decisions. |

### Future Scope

The Student Task Manager System is designed with extensibility in mind. The following enhancements are planned for future iterations:

- Smart Deadline Alerts — Email and browser push notifications for approaching or overdue task deadlines.
- Calendar View — A visual month/week calendar displaying tasks by deadline date for better schedule planning.
- Team Collaboration — Allow students to share tasks or create collaborative group project boards.
- AI-Powered Prioritization — Machine learning model to automatically suggest task priority based on deadlines and workload patterns.
- File Attachments — Allow students to attach reference files (PDFs, images) directly to task entries.
- Analytics Dashboard — Weekly/monthly productivity reports showing task completion rates and productivity trends.
- Mobile App — React Native mobile application for iOS and Android with offline task access.

- Multi-Language Support — Interface localization starting with Bengali for local university users.

## **Conclusion**

The Student Task Manager System addresses a genuine, everyday challenge faced by university students worldwide. By leveraging the power of the MERN stack — one of the most widely adopted full-stack JavaScript ecosystems in modern web development — this project delivers a practical, scalable, and professionally structured solution.

The system not only fulfills its immediate academic objectives but also serves as a meaningful portfolio project demonstrating real-world skills in React component architecture, RESTful API design, JWT-based authentication, MongoDB schema design, and cloud deployment. Every design decision — from the database schema to the UI layout — is grounded in practical software engineering principles rather than academic exercise alone.

The modular architecture ensures that the application can grow beyond its initial scope. With a clear roadmap for future enhancements including AI-powered prioritization, calendar integration, and mobile applications, the Student Task Manager System is positioned as a foundation for a genuinely useful academic productivity tool — one that could meaningfully improve the daily lives of students at IIUC and beyond.